MTH-262 Statistics and Probability Theory

Lecture 09

Review of Lecture 08

- Addition Law of Probability for Mutually Exclusive and Non-Mutually Exclusive Events
- Conditional Probability
- Statistical Independence

$$P(B|A) = P(B),\ P(A|B) = P(A)$$

- Product Law of Probability for dependent events

$$P(A \cap B) = P(A).P(B|A), \qquad \text{provided } P(A)>0$$

- Product Law of Probability for independent events

$$P(A \cap B) = P(A).P(B), \quad \text{provided } P(B|A)=P(B)$$

Topics of This Lecture

Law of Total Probability

&

Bayes Theorem

Law of Total Probability

Suppose that A_1, A_2, …., A_k are mutually exclusive and collectively exhaustive events; then for any event "B",

$$P(B) = \sum_{j=1}^{k} P(B \mid A_j) P(A_j)$$
$$= P(B \mid A_1)P(A_1) + P(B \mid A_2)P(A_2) + \ldots + P(B \mid A_k)P(A_k)$$

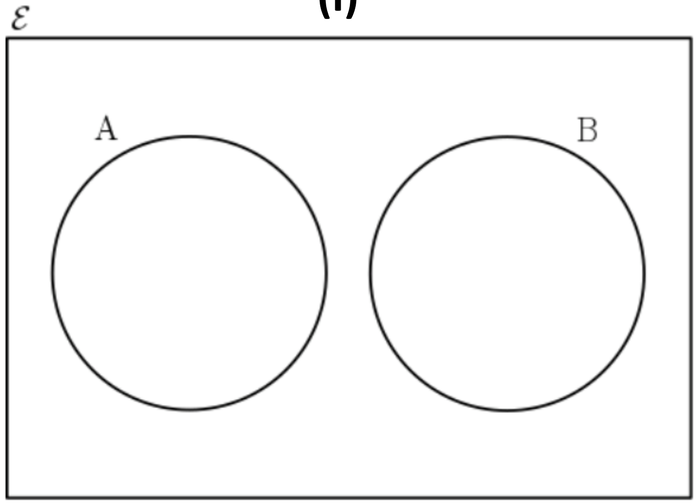
Mutually Exclusive Events: $A_1 \cap A_2 \cap A_3, \ldots, \cap A_k = \emptyset$

Collectively Exhaustive Events: $A_1 \cup A_2 \cup A_3, \ldots, \cup A_k = S$

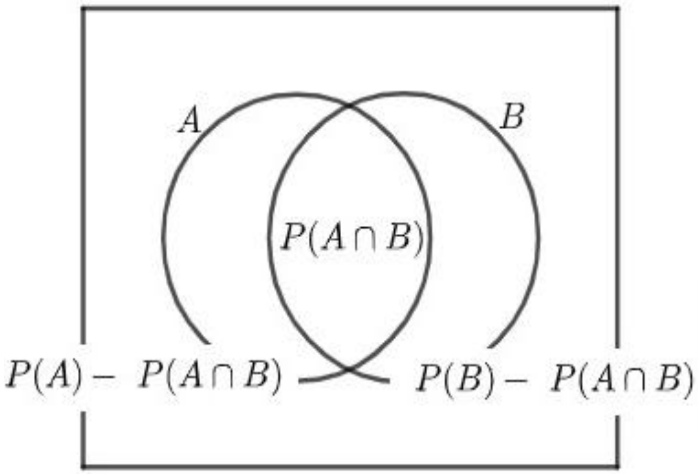
Definitions (Using Venn Diagram)

- Mutually Exclusive Events
- Non-Mutually Exclusive Events
- Collectively Exhaustive Events

(i)



(ii)



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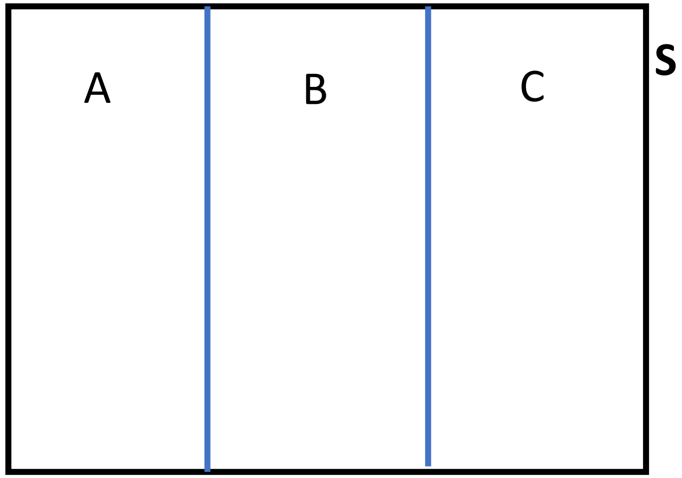
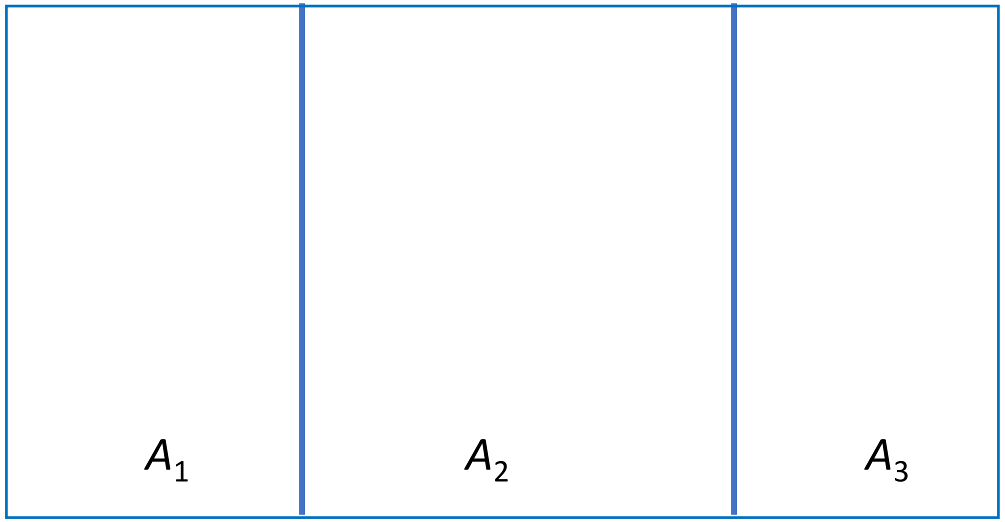


https://studywell.com/as-maths/probability/mutually-exclusive-independent/

Law of Total Probability
(Derivation)

$$P(B) = \sum_{j=1}^{k} P(B \mid A_j)P(A_j)$$
$$= P(B \mid A_1)P(A_1) + P(B \mid A_2)P(A_2) + \ldots + P(B \mid A_k)P(A_k)$$

Consider three events A_1, A_2, A_3,

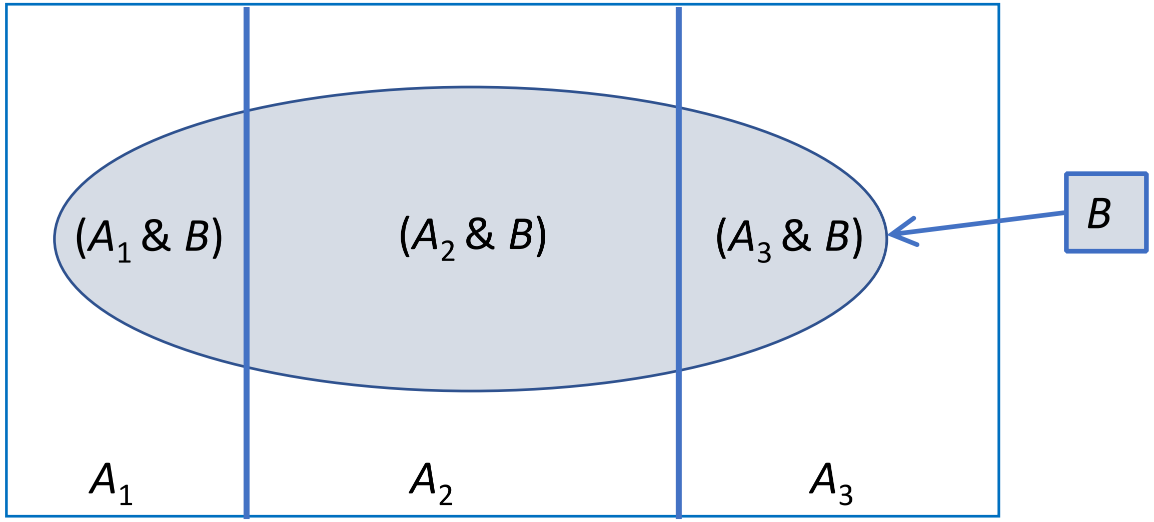


Events A_1, A_2 and A_3 are mutually exclusive and exhaustive events.

Law of Total Probability
(Derivation)

$$P(B) = \sum_{j=1}^{k} P(B \mid A_j)P(A_j)$$
$$= P(B \mid A_1)P(A_1) + P(B \mid A_2)P(A_2) + \ldots + P(B \mid A_k)P(A_k)$$

Consider another event B



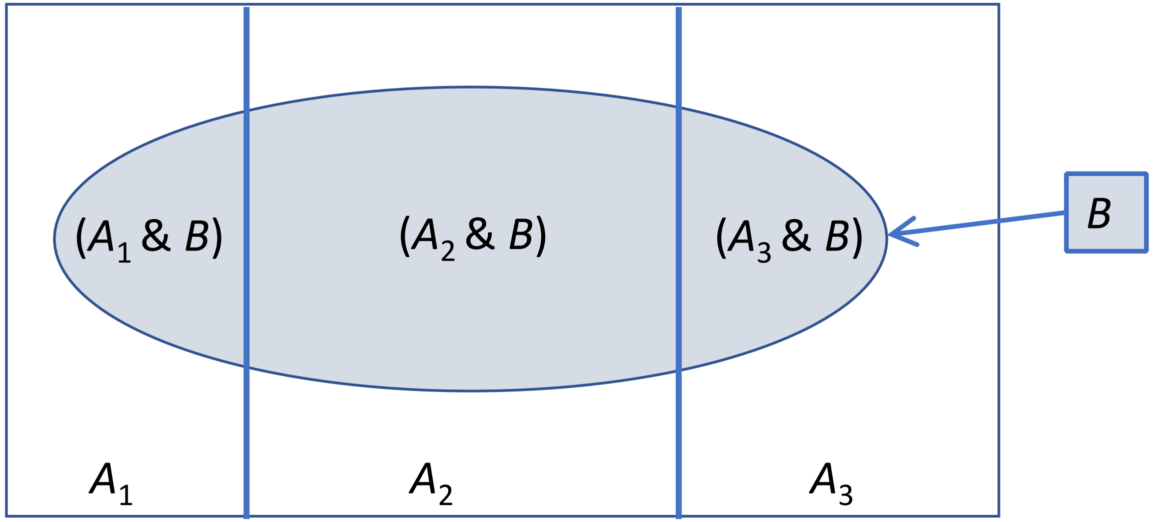
Note that event *B* comprises of three mutually exclusive events (A_1 & B), (A_2 & B) and (A_3 & B).

This shows that if event *B* occurs one of the events A_1, A_2 and A_3 must occur.

Law of Total Probability
(Derivation)

$$P(B) = \sum_{j=1}^{k} P(B \mid A_j)P(A_j)$$
$$= P(B \mid A_1)P(A_1) + P(B \mid A_2)P(A_2) + \ldots + P(B \mid A_k)P(A_k)$$

The Blue colored region



$B = (A_1 \& B) \cup (A_2 \& B) \cup (A_3 \& B)$ Then $P(B) = P(A_1 \& B) + P(A_2 \& B) + P(A_3 \& B)$

or

$P(B) = P(B \mid A_1)\, P(A_1) + P(B \mid A_2)\, P(A_2) + P(B \mid A_3)\, P(A_3)$

Law of Total Probability

Suppose that A_1, A_2, …., A_k are mutually exclusive and collectively exhaustive events; then for any event “B”,

$$P(B) = \sum_{j=1}^{k} P(B \mid A_j) P(A_j)$$
$$= P(B \mid A_1)P(A_1) + P(B \mid A_2)P(A_2) + \ldots + P(B \mid A_k)P(A_k)$$

Mutually Exclusive Events: $A_1 \cap A_2 \cap A_3, \ldots, \cap Ak = \emptyset$

Collectively Exhaustive Events: $A_1 \cup A_2 \cup A_3, \ldots, \cup A_k = S$

Law of Total Probability (Example)

Question: In a certain assembly plant, three machines, B_1, B_2, and B_3, make 30%, 45%, and 25%, respectively, of the products. It is known from past experience that 2%, 3%, and 2% of the products made by each machine, respectively, are defective. Now, suppose that a finished product is randomly selected. What is the probability that it is defective?

Solution:

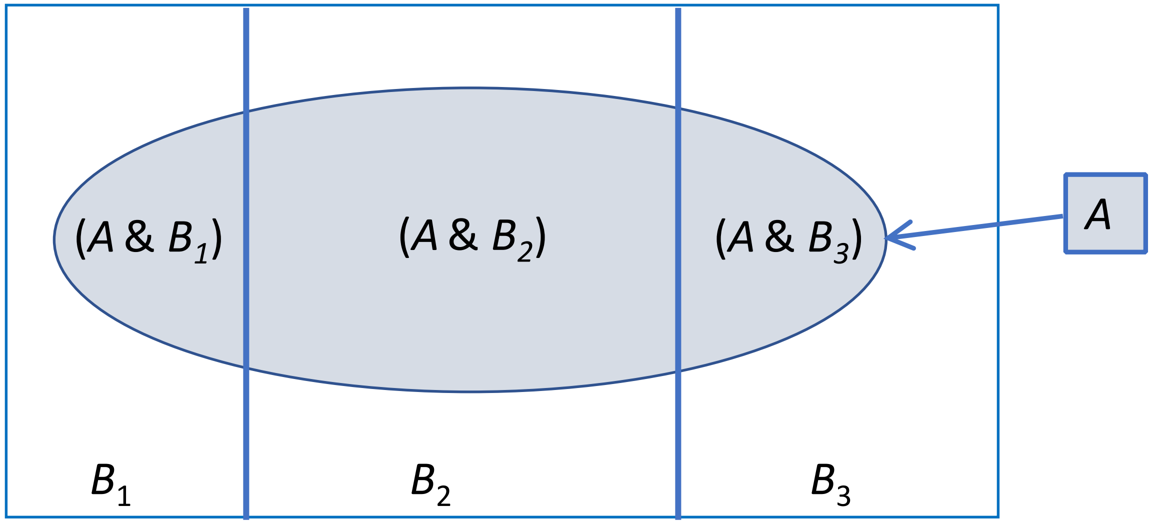
Given Information: B_i = Products produced by Machine i, i=1,2,3. A = Defective products.

$$P(B_1) = 0.30,\ P(B_2) = 0.45,\ P(B_3) = 0.25$$

$$P(A|B_1) = 0.02,\ P(A|B_2) = 0.03,\ P(A|B_3) = 0.02$$

Law of Total Probability (Example)

Solution:



$P(A) = P(B_1)P(A|B_1) + P(B_2)P(A|B_2) + P(B_3)P(A|B_3).$

$P(A) = 0.006 + 0.0135 + 0.005 = 0.0245.$

Conclusion: 2.45 % Chances that the selected product is defective.

Bayes' Theorem

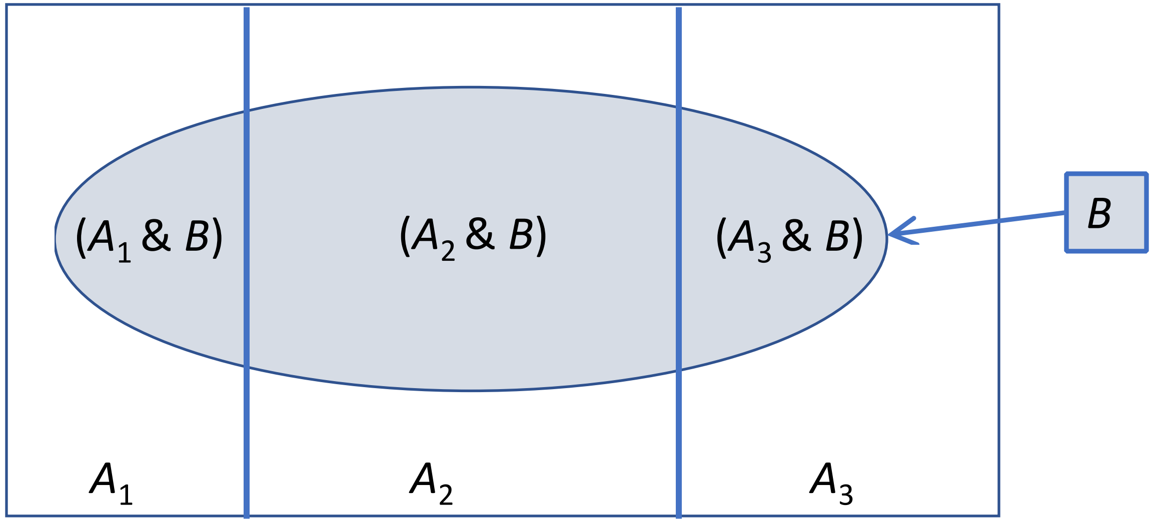
Suppose that A_1, A_2, …., A_k are mutually exclusive and collectively exhaustive events; then for any event "B",

$$P(A_i|\ B) = \frac{P(B|A_i)P(A_i)}{\sum_{j=1}^{k} P(B|A_j)P(A_j)}$$

$$= \frac{P(B|A_i)P(A_i)}{P(B|A_1)P(A_1) + P(B|A_2)P(A_2) + \cdots + P(B|A_k)P(A_k)}$$

where A_{i} can be any one of the events A_1, A_2, …, A_{k}

Bayes' Theorem (Derivation)



Recall conditional probability rule.

$$P(B \mid A) = \frac{P(A \,\&\, B)}{P(A)}$$

So, the probability $P(A_2|B)$ is

$$P(A_2|B) = \frac{P(A_2 \ \&B)}{P(B)} = \frac{P(B \ \&A_2)}{P(B)}$$

From the General multiplication rule we know that

$$P(B \ \&A_2) = P(B|A_2 \ )P(A_2)$$

And the rule of total probability is

$$P(B) = P(B|A_1 \ )P(A_1) + P(B|A_2 \ )P(A_2) + P(B|A_3 \ )P(A_3)$$

Bayes' Theorem (Derivation)

Using the Law of Total probability, We can write the probability as given below:

$$P(A_2|\ B) = \frac{P(B\ \&A_2)}{P(B)}$$

$$= \frac{P(B|A_2\ )P(A_2)}{P(B|A_1\ )P(A_1) + P(B|A_2\ )P(A_2) + P(B|A_3\ )P(A_3)}$$

The same way as $P(A_2|B)$, we can find out $P(A_1|B)$ and $P(A_2|B)$.

Bayes' Theorem (Example)

Question: A manufacturing firm employs three analytical plans for the design and development of a particular product. For cost reasons, all three are used at varying times. In fact, plans 1, 2, and 3 are used for 30%, 20%, and 50% of the products, respectively. The defect rate is different for the three procedures as follows:

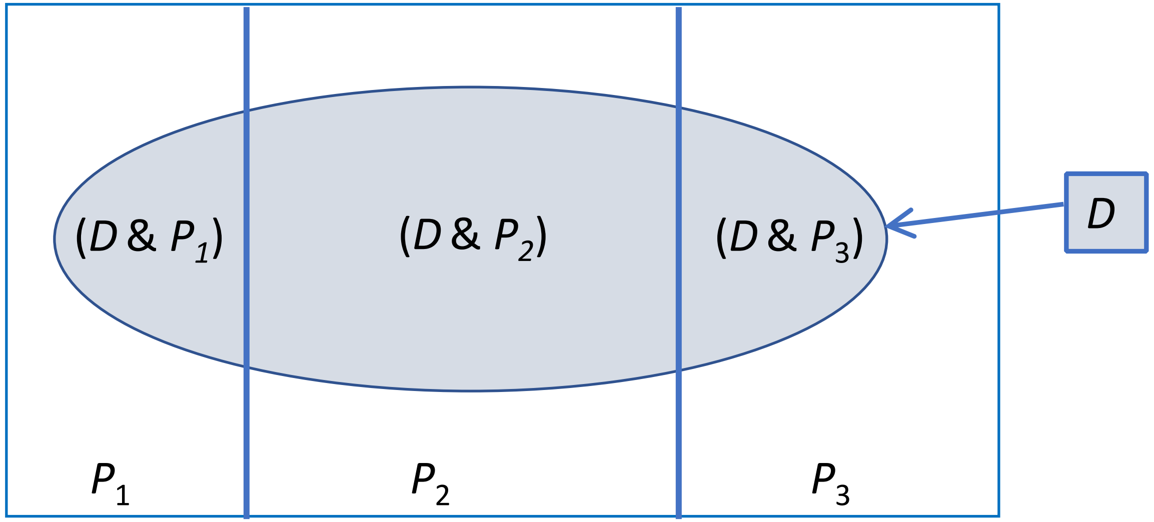
$$P(D|P_1) = 0.01,\ P(D|P_2) = 0.03,\ P(D|P_3) = 0.02,$$

where $P(D|P_j)$ is the probability of a defective product, given plan j. If a random product was observed and found to be defective, which plan was most likely used and thus responsible?

Bayes' Theorem (Example)

From the statement of the problem

$P(P_1) = 0.30$, $P(P_2) = 0.20$, and $P(P_3) = 0.50$,



$$P(P_1|D) = \frac{P(P_1 \cap D)}{P(D)}$$

$$P(P_1|D) = \frac{P(P_1 \cap D)}{\sum_{i=1}^{3} P(P_i).P(D|Pi)}$$

$$P(P_1|D) = \frac{(0.30)*(0.01)}{(0.30)*(0.01)+(0.20)*(0.03)+(0.50)*(0.02)}$$

Some applications
of
Probability

Application 1

A shipment of 7 television sets contains 2 defective sets. A hotel makes a random purchase of 3 of the sets. Find the probability that;

(i) 2 are defective and one non-defective .

(ii) At least one is defective

(iii) All are non-defective

Solution

Given Information: Number of Defectives = 2

Number of Non-Defectives = 5

Total TV = 7

(i) Find the probability that “2 defective and one non-defective are purchased”.

$$P(\text{2 defective and one non-defective}) = \frac{\binom{2}{2}\binom{5}{1}}{\binom{7}{3}} = \frac{5}{35} = 0.143$$

14% are the chances that 2 defective and one non defective TV purchased.

Solution

Given Information: Number of Defectives = 2

Number of Non-Defectives = 5

Total TV = 7

(ii) Find the probability that “At least one is defective”.

$$P(\text{at least one is defective}) = \frac{\binom{2}{1}\binom{5}{2}}{\binom{7}{3}} + \frac{\binom{2}{2}\binom{5}{1}}{\binom{7}{3}} = \frac{25}{35} = 0.714$$

71% are the chances that at least one defective is there in purchased TVs.

Solution

Given Information: Number of Defectives = 2

Number of Non-Defectives = 5

Total TV = 7

(iii) Find the probability that “All are non-defective”.

$$P(\text{all are non-defective}) = \frac{\binom{2}{0}\binom{5}{3}}{\binom{7}{3}} = \frac{10}{35} = 0.29$$

29% are the chances that 2 defective and one non defective TV purchased.

Application 2

John is going to graduate from an industrial engineering department in a university by the end of the semester. After being interviewed at two companies he likes, he assesses that his probability of getting an offer from company A is 0.8, and his probability of getting an offer from company B is 0.6. If he believes that the probability that he will get offers from both companies is 0.5, what is the probability that he will get at least one offer from these two companies?

Solution

Given Information $P(A) = 0.8$

$P(B) = 0.6$

$P(A \cap B) = 0.5$

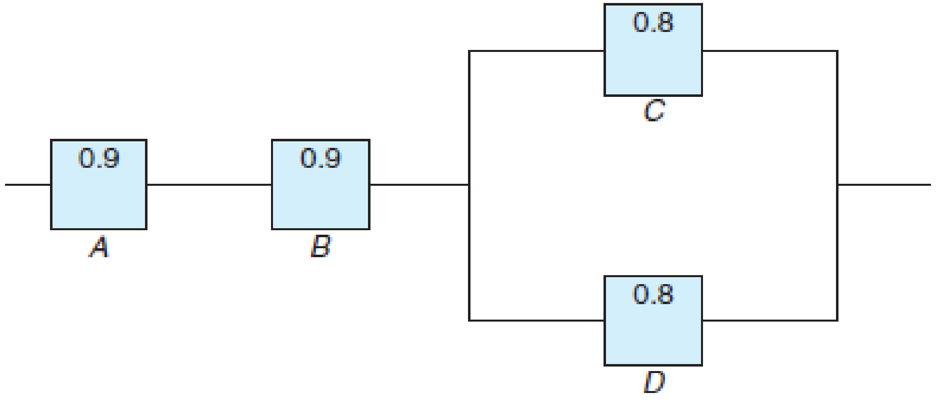
P(at least one offer from these two companies) = $P(A \cup B)$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
$$= 0.8 + 0.6 - 0.5$$
$$= 0.9$$

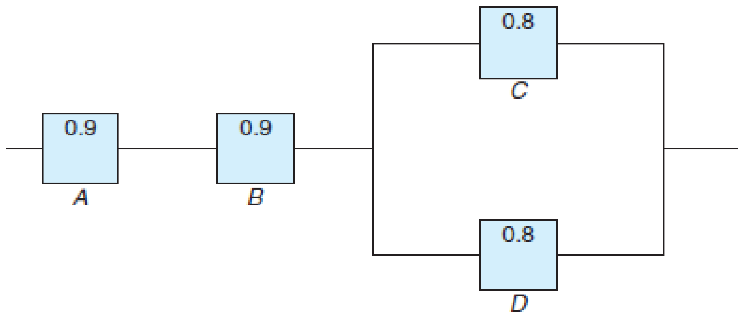
There are 90% chances that the candidate will get at least one offer.

Application 3

An electrical system consists of four components as illustrated in following Figure. The system works if components A and B work and either of the components C or D works. The reliability (probability of working) of each component is also shown in Figure. Find the probability that (a) the entire system works and (b) the component C does not work, given that the entire system works. Assume that the four components work independently.



Solution



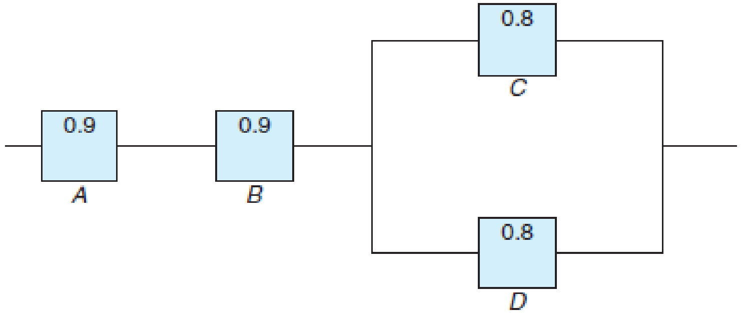
In this configuration of the system, A, B, and the subsystem C and D constitute a serial circuit system, whereas the subsystem C and D itself is a parallel circuit system.

(a) Clearly the probability that the entire system works can be calculated as:

$$P[A \cap B \cap (C \cup D)] = P(A)P(B)P(C \cup D) = P(A)P(B)[1 - P(C' \cap D')]$$
$$= P(A)P(B)[1 - P(C')P(D')]$$
$$= (0.9)(0.9)[1 - (1 - 0.8)(1 - 0.8)] = 0.7776.$$

The equalities above hold because of the independence among the four components.

Solution



(b) To calculate the conditional probability in this case, notice that

$$P = \frac{P(\text{the system works but C doesn't work})}{P(\text{the system works})}$$

$$P = \frac{P(A \cap B \cap C' \cap D)}{P(\text{the system works})}$$

$$P = \frac{0.9 * 0.8 * (1 - 0.8) * 0.8}{0.7776} = 0.1667$$

There are 16.67% chances the component C does not work, given that the entire system works.

Application 4

Three cards are drawn in succession, without replacement, from an ordinary deck of playing cards. Find the probability that the event $A1 \cap A2 \cap A3$ occurs, where $A1$ is the event that the first card is a red ace, $A2$ is the event that the second card is a 10 or a jack, and $A3$ is the event that the third card is greater than 3 but less than 7.

Solution

First, we define the events

A1: the first card is a red ace,

A2: the second card is a 10 or a jack,

A3: the third card is greater than 3 but less than 7.

Now $$P(A_1 \cap A_2 \cap A_3) = P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2)$$

$$= \left(\frac{2}{52}\right)\left(\frac{8}{51}\right)\left(\frac{12}{50}\right) = \left(\frac{8}{5525}\right)$$

Application 5

A paint-store chain produces and sells latex and semigloss paint. Based on long-range sales, the probability that a customer will purchase latex paint is 0.75. Of those that purchase latex paint, 60% also purchase rollers. But only 30% of semigloss paint buyers purchase rollers. A randomly selected buyer purchases a roller and a can of paint. What is the probability that the paint is latex?

Solution

Let's define events: L = Customer will purchase Latex paint

S = Customer will purchase Semigloss paint

R = Paint buyers purchase rollers

P(L) = 0.75, P(S) = 0.25

P(R|L) = 0.60, P(R|S) = 0.30

We have to find the probability that paint purchased paint is Latex given that the customer bought the roller.

$$P(L|R) = \frac{(0.75*0.60)}{(0.75*0.60) + (0.25*0.30)}$$

Applications

2.76 In an experiment to study the relationship of hypertension and smoking habits, the following data are collected for 180 individuals:

	Nonsmokers	Moderate Smokers	Heavy Smokers
H	21	36	30
NH	48	26	19

where H and NH in the table stand for *Hypertension* and *Nonhypertension*, respectively. If one of these individuals is selected at random, find the probability that the person is

(a) experiencing hypertension, given that the person is a heavy smoker;

(b) a nonsmoker, given that the person is experiencing no hypertension.

2.88 Before the distribution of certain statistical software, every fourth compact disk (CD) is tested for accuracy. The testing process consists of running four independent programs and checking the results. The failure rates for the four testing programs are, respectively, 0.01, 0.03, 0.02, and 0.01.

(a) What is the probability that a CD was tested and failed any test?

(b) Given that a CD was tested, what is the probability that it failed program 2 or 3?

(c) In a sample of 100, how many CDs would you expect to be rejected?

(d) Given that a CD was defective, what is the probability that it was tested?

Applications

2.101 A paint-store chain produces and sells latex and semigloss paint. Based on long-range sales, the probability that a customer will purchase latex paint is 0.75. Of those that purchase latex paint, 60% also purchase rollers. But only 30% of semigloss paint buyers purchase rollers. A randomly selected buyer purchases a roller and a can of paint. What is the probability that the paint is latex?

2.114 A shipment of 12 television sets contains 3 defective sets. In how many ways can a hotel purchase 5 of these sets and receive at least 2 of the defective sets?

2.118 A certain form of cancer is known to be found in women over 60 with probability 0.07. A blood test exists for the detection of the disease, but the test is not infallible. In fact, it is known that 10% of the time the test gives a false negative (i.e., the test incorrectly gives a negative result) and 5% of the time the test gives a false positive (i.e., incorrectly gives a positive result). If a woman over 60 is known to have taken the test and received a favorable (i.e., negative) result, what is the probability that she has the disease?